

Toxic Comment Severity Ranking Using Machine Learning

Project Report

Author

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Project Type

Natural Language Processing (NLP) and Machine Learning

Dataset

Kaggle Jigsaw Toxic Severity Rating Competition

Tools and Technologies

Python, Pandas, NumPy, Scikit-learn, Matplotlib, Jupyter Notebook

1. Introduction

Online communities generate large amounts of user-generated content every day. While these platforms enable communication and collaboration, they are also vulnerable to abusive, offensive, and toxic comments. Manual moderation is time-consuming and difficult to scale.

The objective of this project is to build a machine learning system capable of ranking comments according to their toxicity severity. Instead of classifying comments as toxic or non-toxic, the model assigns scores such that more toxic comments receive higher scores than less toxic comments.

This project uses the Kaggle Jigsaw Toxic Severity Rating dataset and applies Natural Language Processing (NLP) techniques combined with machine learning algorithms to solve the ranking problem.

2. Problem Statement

The Jigsaw Toxic Severity Rating competition focuses on ranking comment pairs.

For each pair:

- less_toxic
- more_toxic

the model must assign a higher toxicity score to the more toxic comment.

The evaluation metric is Ranking Accuracy, which measures the percentage of comment pairs that are correctly ordered.

A ranking is considered correct when:

$\text{Score}(\text{more_toxic}) > \text{Score}(\text{less_toxic})$

3. Dataset Description

The project uses two datasets:

comments_to_score.csv

Contains comments that require toxicity severity scores.

validation_data.csv

Contains pairs of comments:

- less_toxic
- more_toxic

These pairs are used to evaluate ranking performance.

Dataset Characteristics

- Thousands of online comments
- Text-based data
- Pairwise toxicity comparisons
- Real-world user-generated content

4. Exploratory Data Analysis (EDA)

Several exploratory analyses were performed before model training.

Comment Length Analysis

Character lengths were calculated for all comments to understand text distribution.

Word Count Analysis

Word counts were analyzed to identify:

- Very short comments
- Very long comments
- Average comment size

Visualizations

The following visualizations were generated:

- Comment Length Distribution
- Word Count Distribution
- Word Count Boxplot

These analyses helped understand the structure and variability of the dataset.

5. Text Preprocessing

Raw text often contains inconsistencies that can negatively impact model performance.

The following preprocessing steps were applied:

Missing Value Handling

Missing comments were replaced with empty strings.

Lowercase Conversion

All text was converted to lowercase to ensure consistency.

Whitespace Normalization

Extra spaces and formatting inconsistencies were removed.

Final Text Corpus

The cleaned text corpus was used for feature engineering and model training.

6. Feature Engineering

Machine learning algorithms cannot directly process raw text.

TF-IDF (Term Frequency–Inverse Document Frequency) vectorization was used to transform text into numerical features.

TF-IDF Configuration

- Maximum Features: 50,000
- N-Gram Range: (1,3)
- Minimum Document Frequency: 2
- Maximum Document Frequency: 0.95
- Sublinear TF Scaling Enabled

Benefits

TF-IDF helps identify important words while reducing the influence of extremely common terms.

7. Model Development

Multiple machine learning models were evaluated.

Models Evaluated

1. Logistic Regression
2. Linear Support Vector Classifier (Linear SVC)
3. Ridge Classifier
4. Passive Aggressive Classifier
5. SGD Classifier

Training Strategy

The validation pairs were converted into a pairwise training dataset.

For each pair:

- less_toxic $\rightarrow$ label 0
- more_toxic $\rightarrow$ label 1

This transformed the ranking problem into a supervised learning task.

8. Model Evaluation

Each model was evaluated using Ranking Accuracy.

Evaluation Process

1. Score validation comments.
2. Compare toxicity scores for each pair.
3. Count correctly ranked pairs.
4. Compute ranking accuracy.

Model Comparison Results

Model	Ranking Accuracy
Linear SVC	80.19%
Ridge	79.63%
Logistic Regression	78.21%
Passive Aggressive	77.96%
SGD	75.73%

Best Performing Model

Linear SVC achieved the highest ranking accuracy and was selected as the final model.

9. Model Interpretation

One advantage of linear models is interpretability.

The coefficients learned by Linear SVC were analyzed to identify:

Top Toxic Words

Words with large positive coefficients increase toxicity scores.

Top Non-Toxic Words

Words with large negative coefficients decrease toxicity scores.

This analysis provides insights into the linguistic patterns learned by the model.

10. Results

Best Model

Linear SVC

Ranking Accuracy

80.19%

Correct Rankings

24,143

Total Validation Pairs

30,108

Vocabulary Size

50,000 TF-IDF Features

Outcome

The model successfully learned toxicity patterns and correctly ranked the majority of validation pairs.

11. Project Strengths

Effective Text Representation

TF-IDF successfully transformed textual comments into meaningful numerical features.

Strong Ranking Performance

The model achieved over 80% ranking accuracy.

Multiple Model Comparison

Several machine learning algorithms were evaluated to identify the best performer.

Model Interpretability

Feature importance analysis provided transparency into model decisions.

12. Limitations

Although the model performed well, several limitations remain.

Context Awareness

TF-IDF cannot fully understand context.

Sarcasm Detection

Sarcastic comments may be difficult to identify.

Semantic Understanding

Words are treated primarily as independent features.

Deep Context

The model lacks the contextual understanding provided by transformer architectures.

13. Conclusion

This project developed an end-to-end toxicity severity ranking system using Natural Language Processing and Machine Learning.

Text comments were transformed using TF-IDF vectorization and multiple ranking models were trained and evaluated on pairwise toxicity comparisons from the Jigsaw Toxic Severity Rating dataset.

Among the evaluated models, Linear SVC achieved the highest ranking accuracy of 80.19%.

The results demonstrate that linear models are highly effective for high-dimensional sparse TF-IDF text representations and can successfully rank comments according to toxicity severity.

The project successfully demonstrates the complete machine learning workflow, including data loading, exploratory analysis, preprocessing, feature engineering, model comparison, ranking evaluation, interpretation, and submission file generation.